

Option decision (LSM vs market) · bar charts

Friday 17 March 2023 monthly expiry · 1-hour lookback · SPY / AAPL / MSFT

What this PDF is for

- This PDF is the bar-chart companion to the option-decision tables for 17 March 2023.
- Each page is a 2×2 of 1-day vs 5-day and hourly vs minutely rolling.
- Four bars per panel: GBM, Merton, GARCH, GARCH-Merton.
- Heston and Heston-Merton are excluded from every chart. Their LSM RMSE is an order of magnitude larger and would flatten the other bars; see the tables PDF for those numbers.
- The outlined bar is the lowest RMSE or MAE on that panel.

Methodology

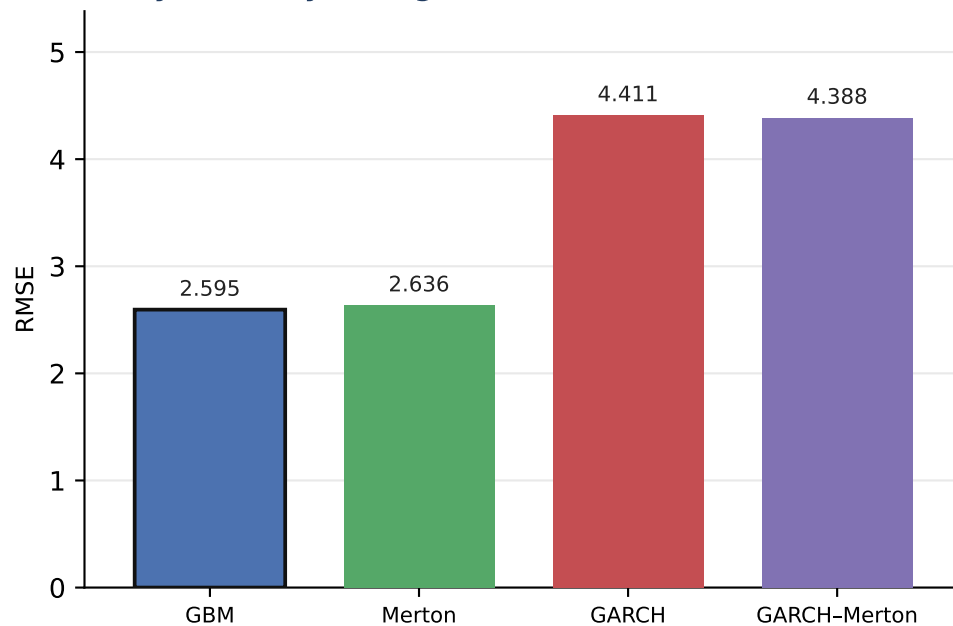
- Calibration and rolling are the same as the stock-path study (1-hour lookback; hourly and minutely; 1-day and 5-day windows).
- Contracts are 12 listed American calls per name from the short-interval options panel, sampled on the study window (seed 42).
- The sample is OTM / ATM / ITM × DTE 7–60 trading days, with unique 1-minute RTH quotes.
- Each contract is priced with Longstaff-Schwartz LSM: 500 risk-neutral paths, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$.
- RMSE and MAE are computed on (model LSM price – listed market price) across the 12 contracts.
- The same seeds and the same 12 contracts are reused across models, including Heston.
- Models priced here: GBM, Merton, Heston, Heston-Merton, GARCH(1,1), and GARCH-Merton.

Heston and Heston-Merton are in the tables PDF only.

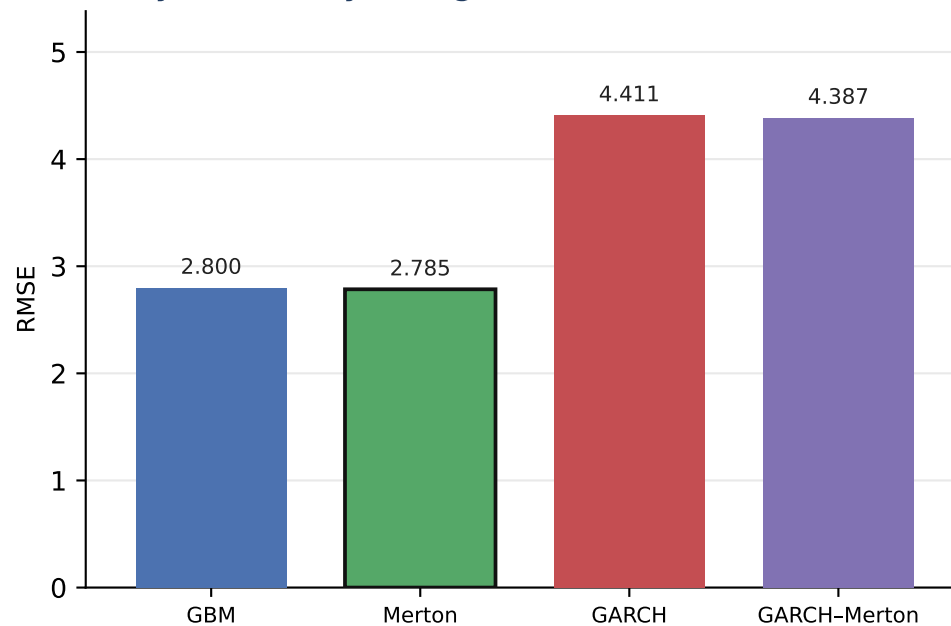
Windows: 1-day = Fri 17 Mar 2023 09:30–15:59 ET (RTH). 5-day = Mon 13 Mar – Fri 17 Mar 2023 (five RTH sessions).

Option RMSE (LSM vs market) · mean of SPY / AAPL / MSFT

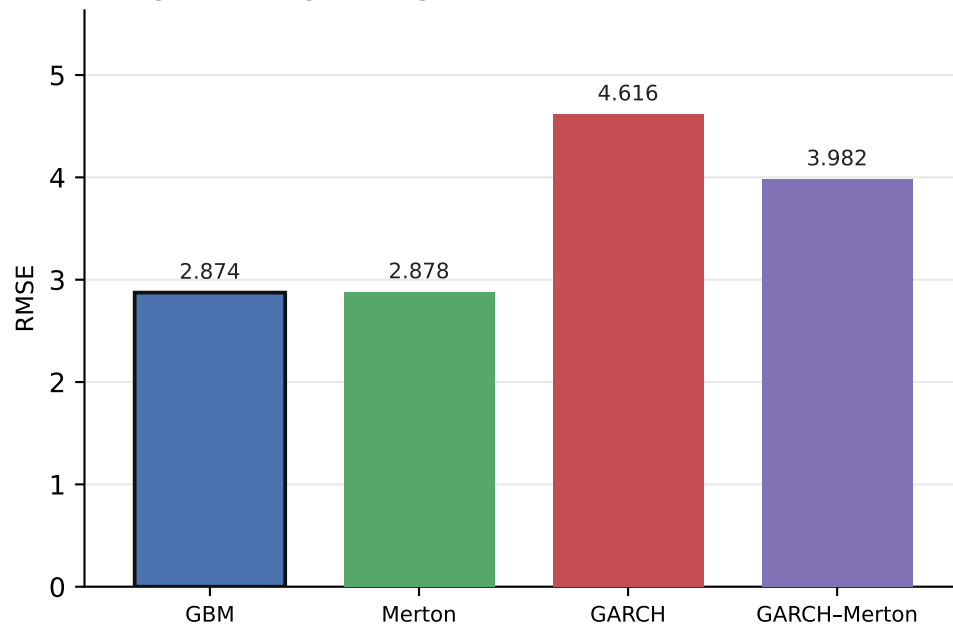
1-day · hourly rolling



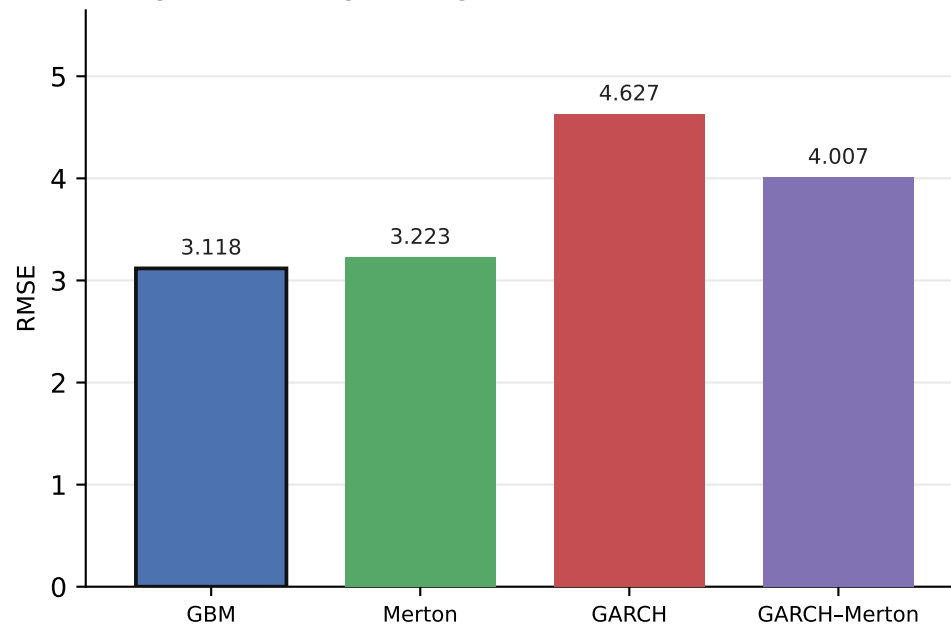
1-day · minutely rolling



5-day · hourly rolling



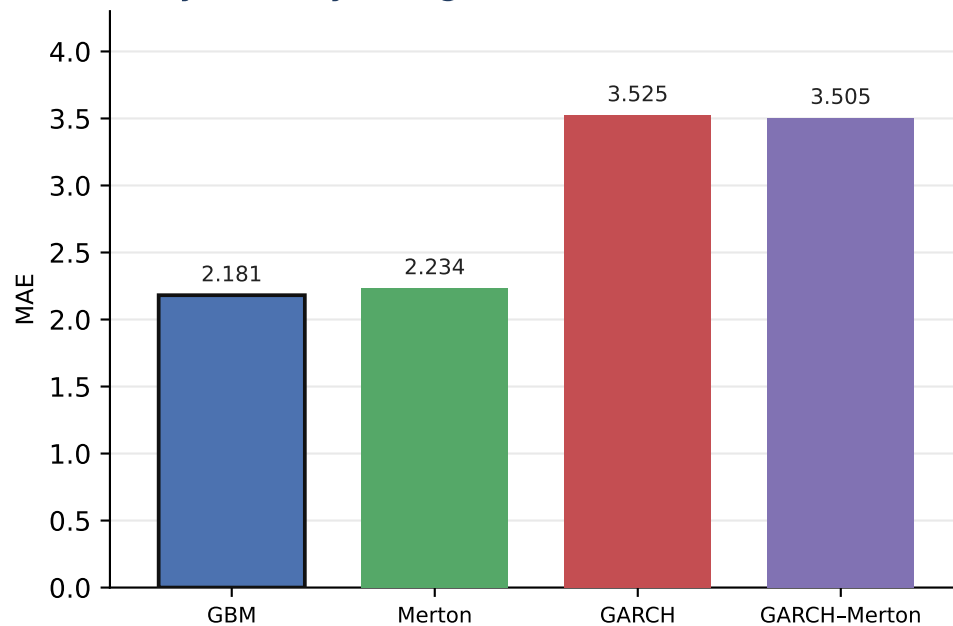
5-day · minutely rolling



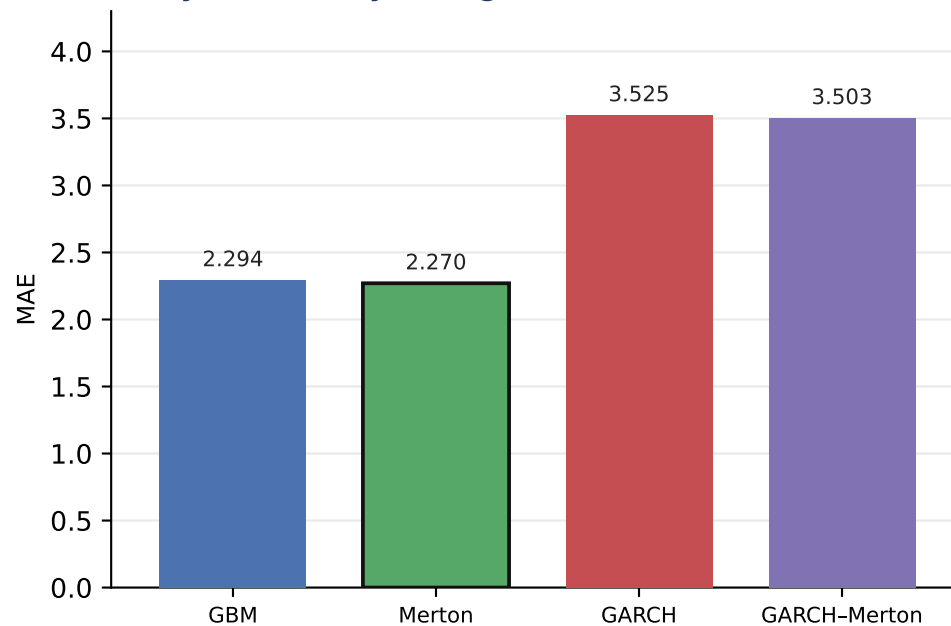
Lowest bar outlined · Heston and Heston-Merton omitted (see tables PDF).

Option MAE (LSM vs market) · mean of SPY / AAPL / MSFT

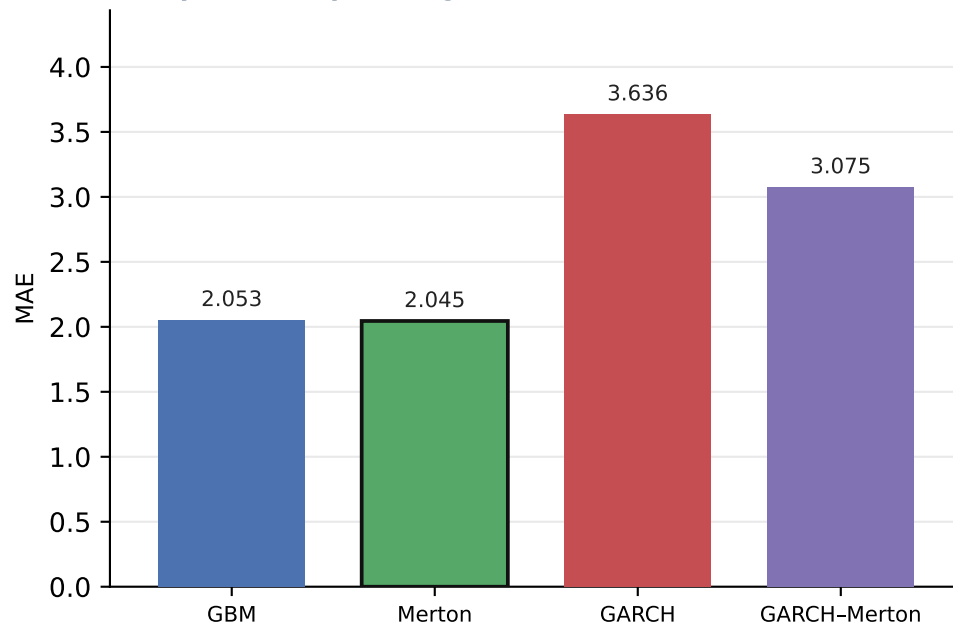
1-day · hourly rolling



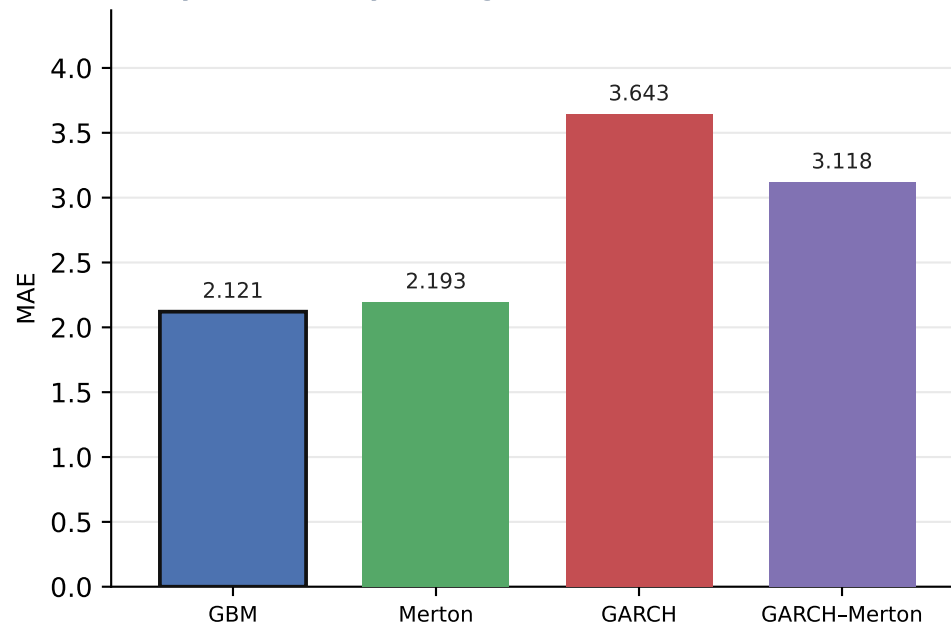
1-day · minutely rolling



5-day · hourly rolling



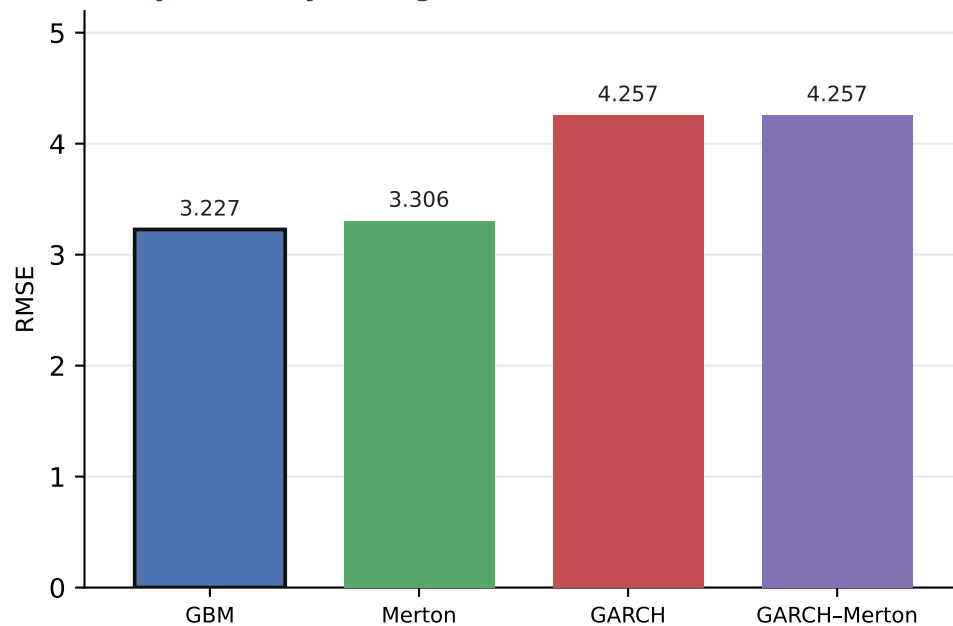
5-day · minutely rolling



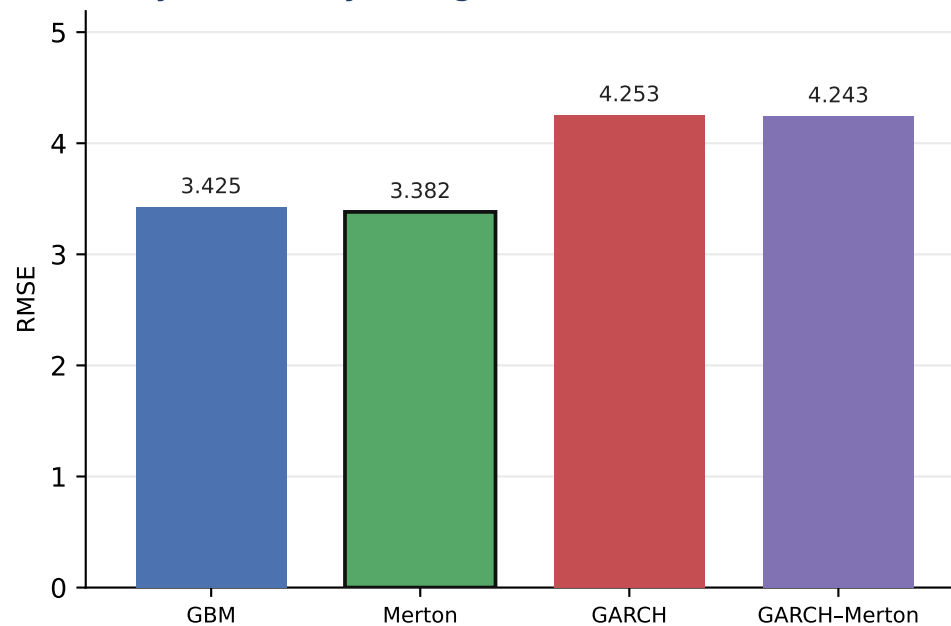
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Option RMSE (LSM vs market) · SPY

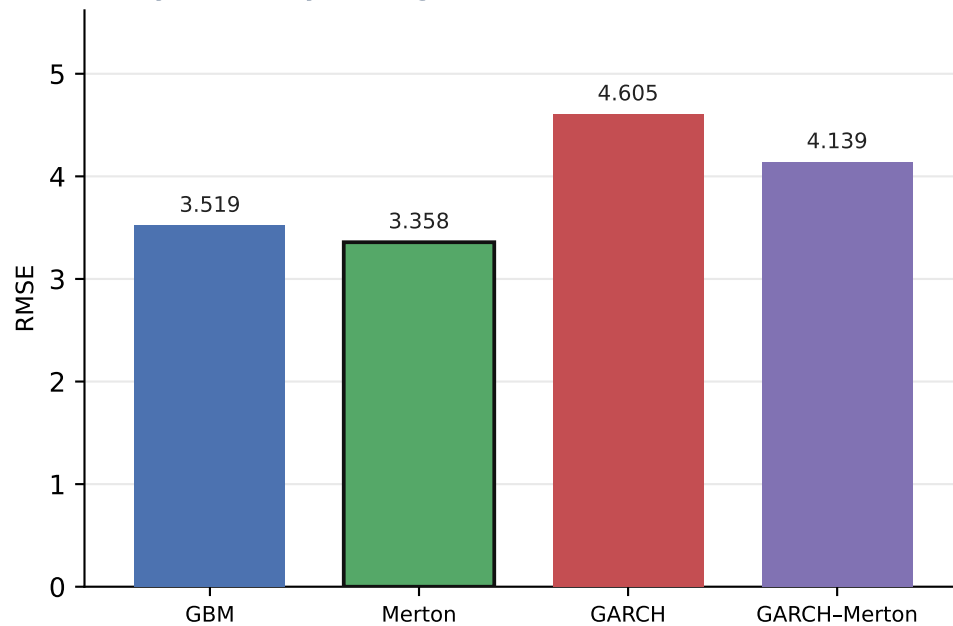
1-day · hourly rolling



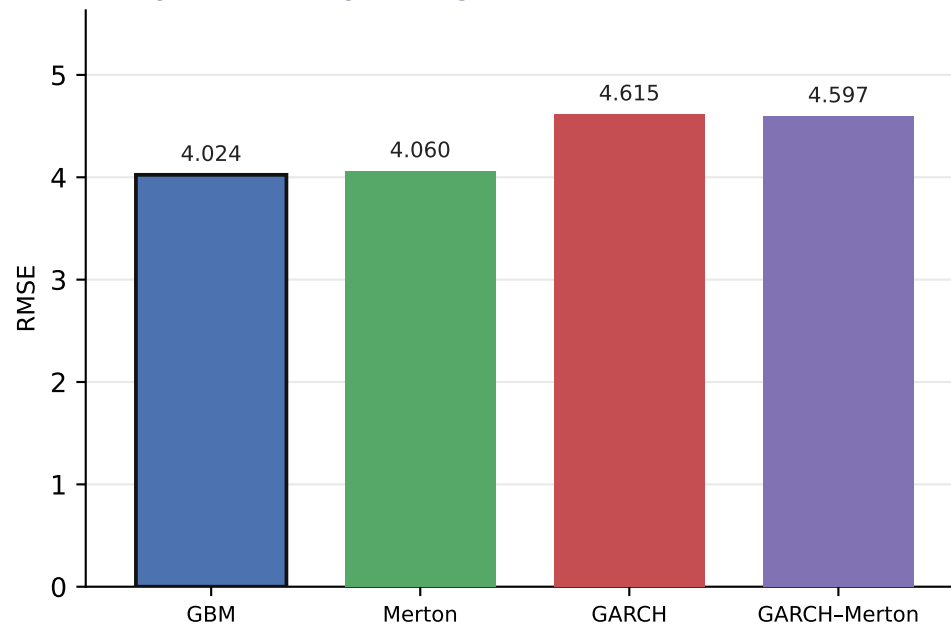
1-day · minutely rolling



5-day · hourly rolling

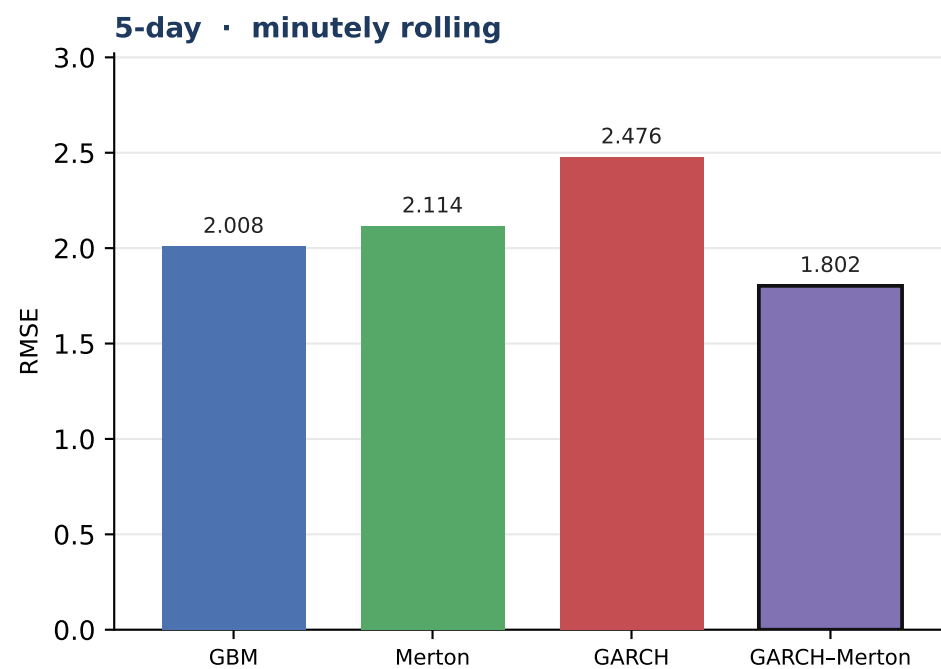
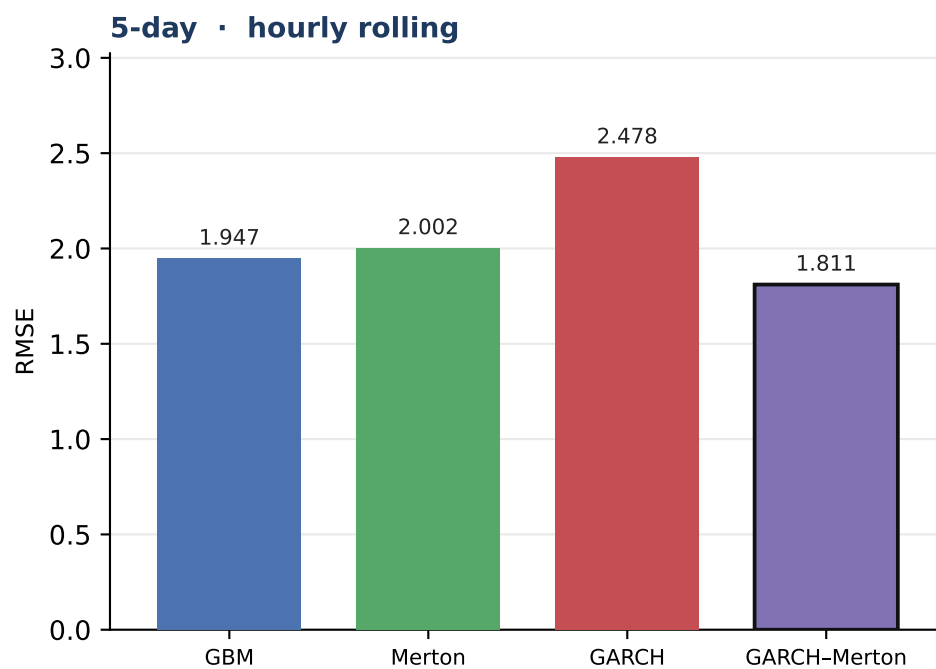
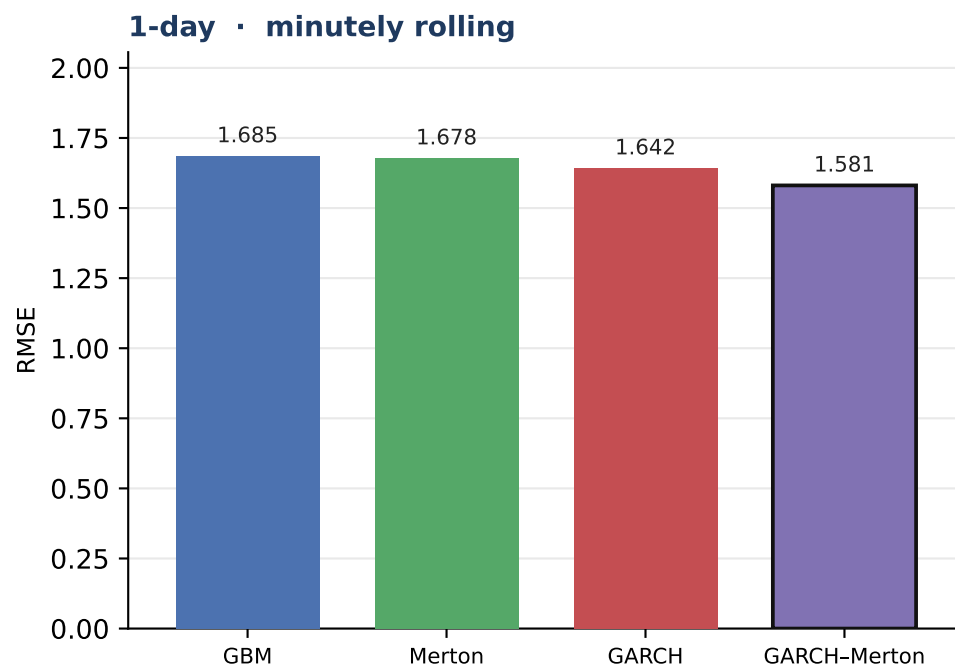
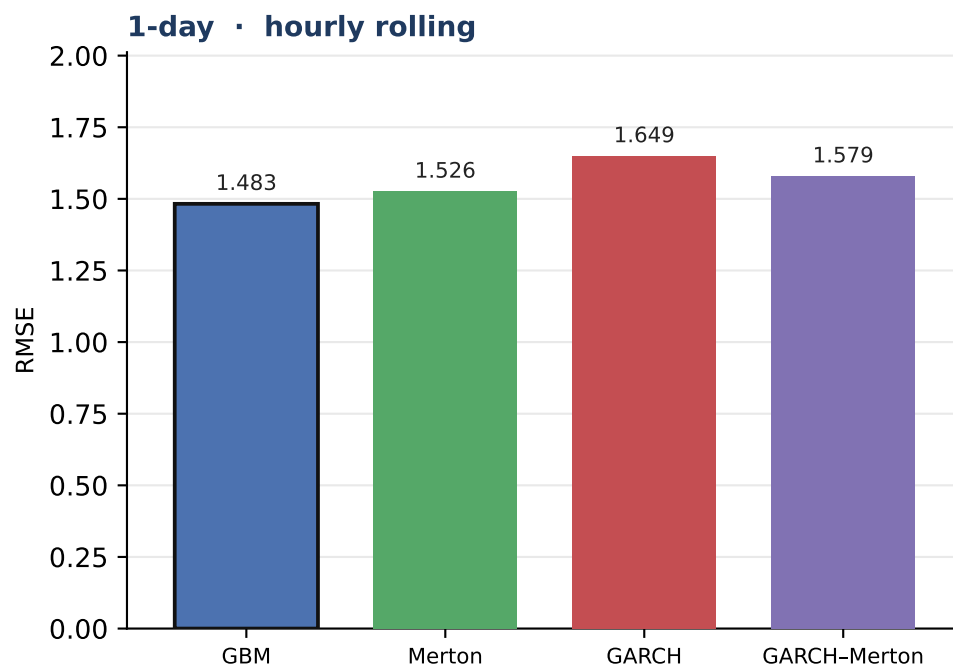


5-day · minutely rolling



Lowest bar outlined · Heston and Heston-Merton omitted (see tables PDF).

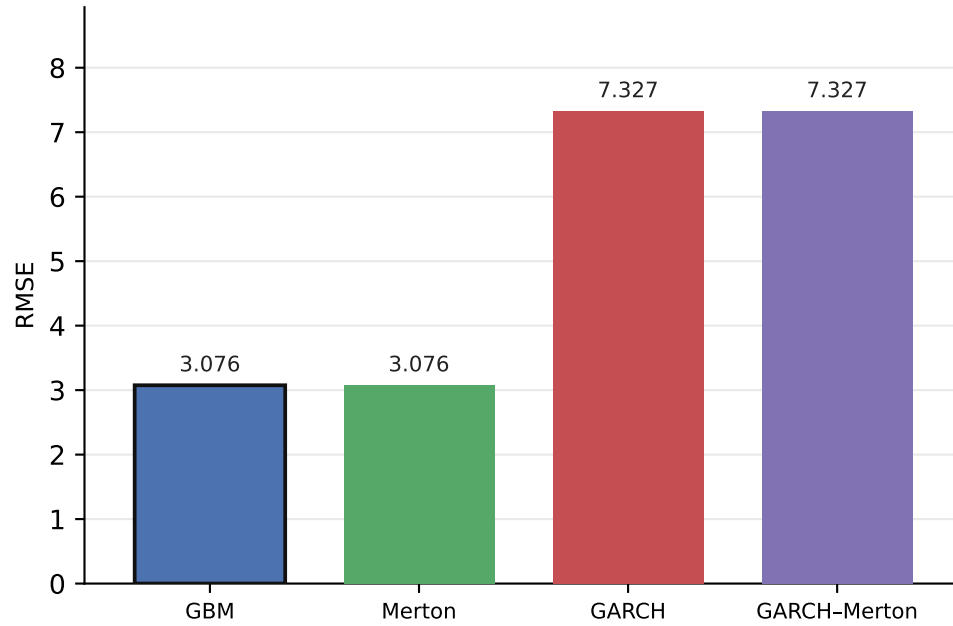
Option RMSE (LSM vs market) · AAPL



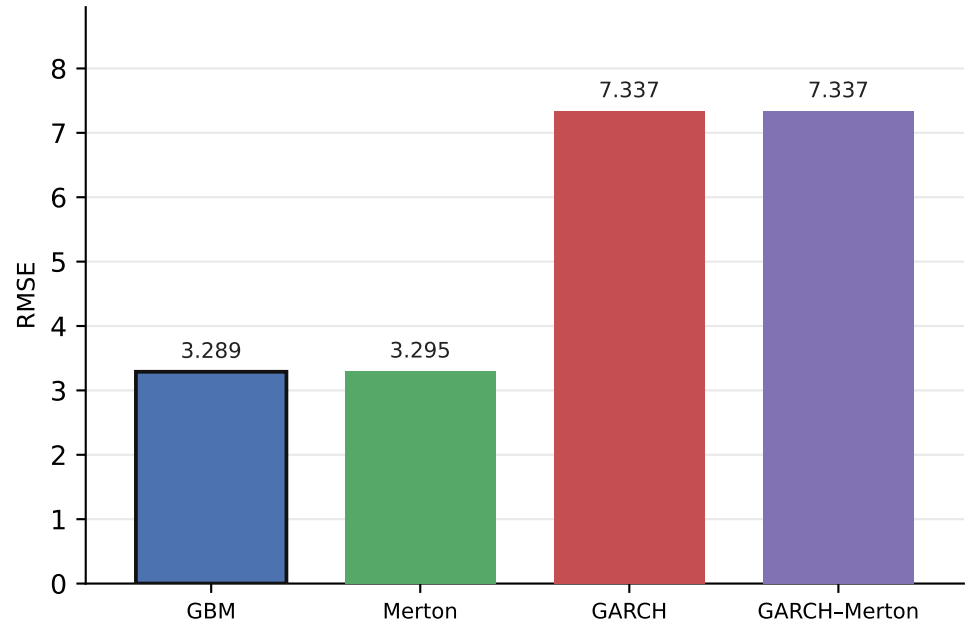
Lowest bar outlined · Heston and Heston-Merton omitted (see tables PDF).

Option RMSE (LSM vs market) · MSFT

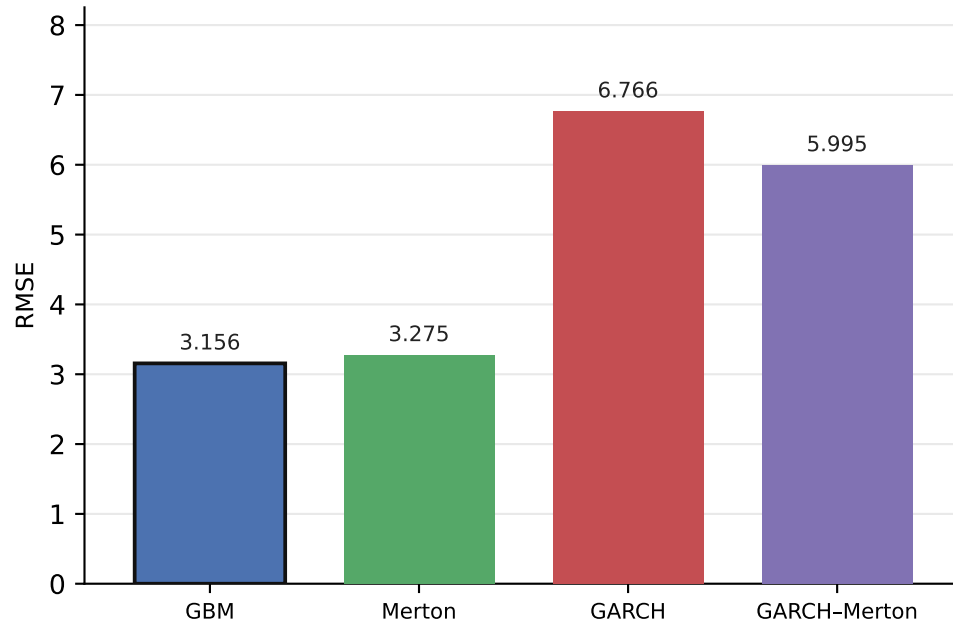
1-day · hourly rolling



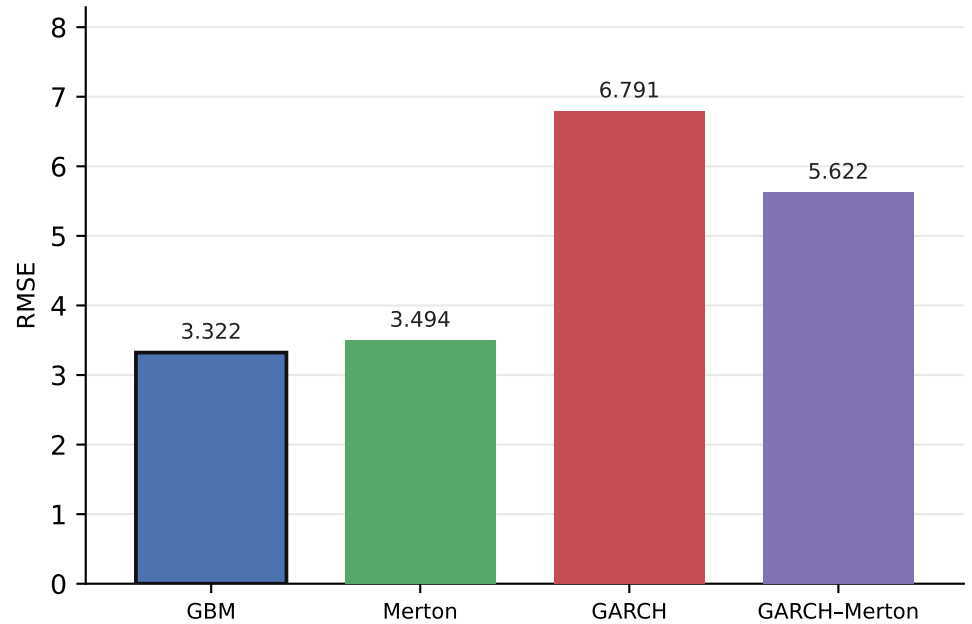
1-day · minutely rolling



5-day · hourly rolling



5-day · minutely rolling



Lowest bar outlined · Heston and Heston-Merton omitted (see tables PDF).